



VIET NAM OIL AND GAS GROUP
BIEN DONG PETROLEUM OPERATING COMPANY

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SAFETY LEVEL TRANSMITTER (DP) WITH DIAPHRAGM SEAL MAINTENANCE INSTRUCTION



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1.0 PURPOSE

This document outlines steps, which are applied to do Maintenance and Function test for Safety Level Transmitter (DP) with Diaphragm Seal within facilities of Bien Dong Petroleum Operating Company. This instruction is to complement and be used in conjunction with:

- Work Order Implementation Procedure BD-OPS-MNR-P-1023
- Maintenance Planning and Scheduling BD-OPS-MNR-P-1022
- Safety Instrument Maintenance / Function Test BD-OPS-MNR-P-3304

The Objectives in this instruction are as follows:

- Provide details guideline/ job steps, which are performed when maintenance and function test of Safety Level Transmitter (DP) with Diaphragm Seal.
- Provide maintenance tasks, based on working condition, and type of devices, and test/ Function test report to record.

2.0 SCOPE

This instruction is applicable corrective or breakdown maintenance and function test Safety Level Transmitter (DP) with Diaphragm Seal on WHP-MT1, WHP-HT1, and PQP-HT of Bien Dong POC.

Work Order Execution, Labor Reporting and Closing Work Order are out of Scope of this instruction. E&I technician will apply the procedure together with Related Work Order Procedure.

This Maintenance Instruction is applied to Level Transmitter (DP) Model: EJX110A with diaphragm seal of Yokogawa and Model: 3051CD of Rosemount

3.0 REFERENCES

- | | |
|--|-------------------|
| • ISO 14001:2015 Standard | |
| • OHSAS 18001:2007 Standard | |
| • Work Order Implementation Procedure: | BD-OPS-MNR-P-1023 |
| • Maintenance Planning and Scheduling: | BD-OPS-MNR-P-1022 |
| • Safety Instrument Maintenance / Function Test: | BD-OPS-MNR-P-3304 |
| • Safety and Environment Management System: | BD-HSE-X-P-0001 |
| • Permit To Work: | BD-HSE-P-0006 |
| • Job Safety Analysis: | BD-HSE-P-0024 |
| • Risk Assessment and Risk Management: | BD-HSE-P-0001 |
| • LO/TO procedures | BD-HSE-X-P-0902 |

4.0 DEFINITIONS AND ABBREVIATION

CCR	Central Control Room
CRE	Control Room Engineer
MOS	Maintenance Override Switch
MT1	Moc Tinh 1 Well Head Platform
HT1	Hai Thach 1 Well Head Platform
PQP-HT	Process and Quarter Platform
HART	Highway Addressable Remote Transducer
H	High Alarm
HH	High High Alarm
LL	Low Low Alarm

LZT Safety Level Transmitter

5.0 SAFETY AND/OR ENVIRONMENTAL REQUIREMENTS

Every endeavor will be made to provide appropriate safety training for all employees, together with any safety equipment and protective clothing. It is also our policy to provide regular safety instruction and to stimulate joint consultation in the area of safety and health.

6.0 INSTRUCTION**6.1 Pre-Works Instruction**

- a. Refer to Safety Instrument Maintenance / Function Test Procedure
- b. Refer to YOKOGAWA transmitter's manual (Type DP, model: EJX110A), ROSEMOUNT transmitter's manual, Model: 3051CD.
- c. Double check and confirm with CRE for MOS is applied prior to start work
- d. Tools preparation;
 - Hand pump with fitting connection.
 - Multi meter Fluke 789.
 - Hart 475

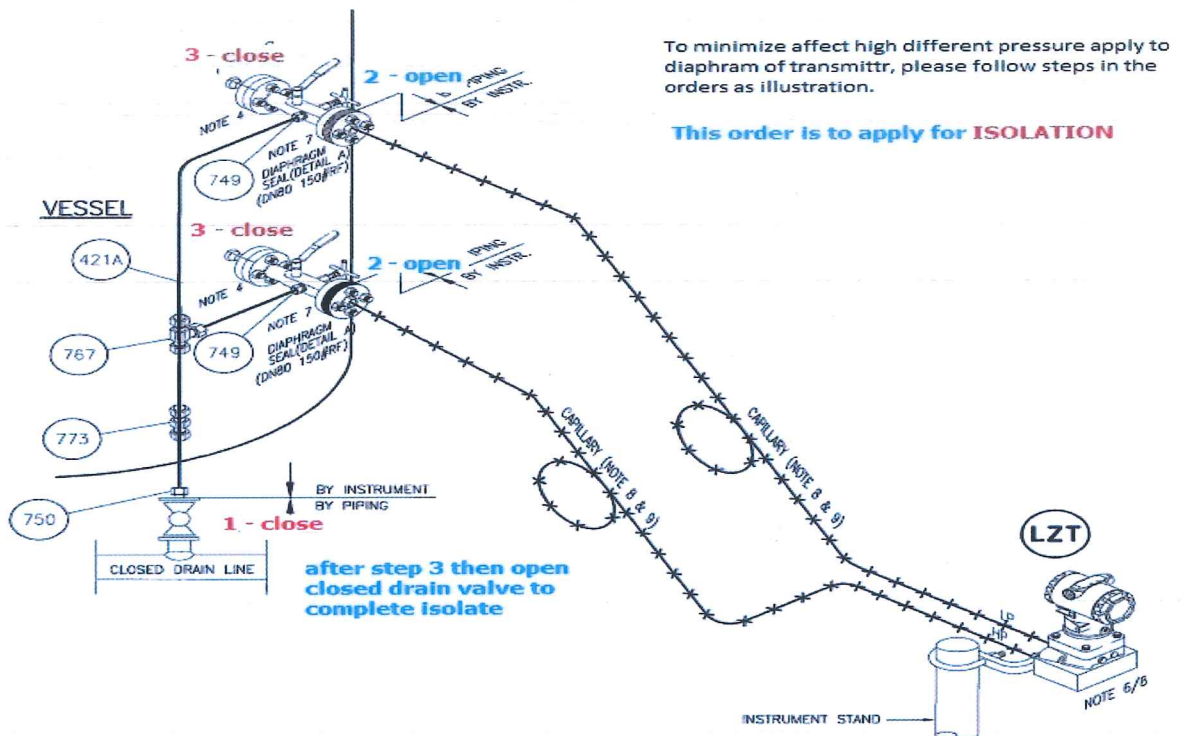
6.2 Works Instruction**6.2.1 Inspection**

- a. Visually inspect the level transmitter (DP) loop components for any signs of damage, corrosion or contamination. Check mountings for security.
- b. Check cables and glands for security/damage and ensure identification is secure, correct and in place.
- c. Internally examine instrument enclosure for signs of moisture ingress. Check all terminations for security/damage and signs of overheating. Ensure all gaskets/seals are clean and intact
- d. Carry out a visual inspection of all instrument sample valve for damage, leakage, corrosion, support and ensure correct installation of fittings. Check there is no intermix of manufacturer's products.
- e. Check tagname.
- f. Ex rating inspection:
 - Equipment is appropriate to area classification.
 - Equipment gas group is correct.
 - Equipment temperature class is correct.
 - Glands certification.
 - There are no unauthorized modifications.
 - Bolts, glands, drain/entry plugs are completed.

- Electrical installation is clean and dry.

6.2.2 Maintenance

Ensure transmitter on MOS Mode from CRE. Apply ISOLATION as figure below



- Visually check Capillary Seal for High Port and Low Port for any signs of damage, corrosion or contamination, clean them if dirty.
- Use HART Communicator connects via HART mode, review configured data such as Equipment Tag, Equipment Description, Sensor Limits, Calibration Range etc.
- Expose the Capillary Seal of the Low Port and High Port to the atmosphere.
- Check the reading value on LCD transmitter, hart display. The process value should be 0%. If process value different 0%, use Hart 475 Trim Zero in auto mode pressure calibration.
- Connect High port to pump. Verify the linearity by increasing and decreasing the pressure (0%, 25%, 50%, 75%, 100%, 75%, 50%, 25% and 0% of range), check comparison % output results between hart display and CCR make sure its correct.
- If error of transmitter greater than accuracy of spec ($\pm 0.5\%$ of reading). Re-calibration will take place as sequences below.

Percent error formula of transmitter:

$$\text{Actual measured (mA)} = (20 - 4) * (\text{Value} - \text{LRV}) * 100\% / (\text{URV} - \text{LRV}) + 4\text{mA}$$

The measuring range of 50 to 250 kPa; the actual measured value of 130 kPa.

$$\text{Actual measured} = (20 - 4)(130 - 50) / (250 - 50) \times 100 + 4 = 10.4\text{mA}$$

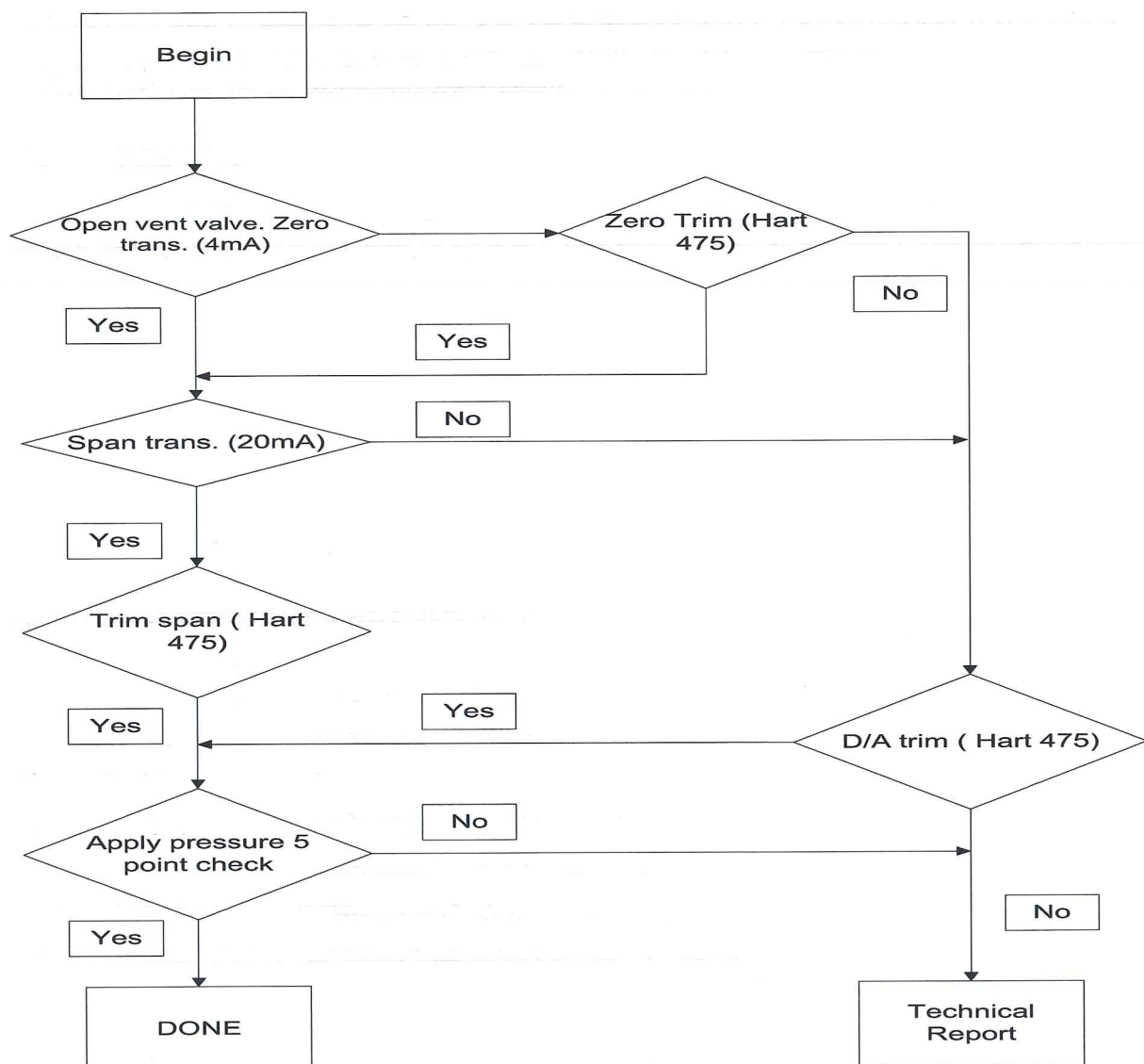
Use HART 475 to check reading value, Connect High port to pump, free port low. Verify the linearity by increasing and decreasing the pressure (0%,25%,50%,75%,100%,75%,50%,25% and 0% of range) and check and compare the accuracy of results % output between Hart display and CCR. Put the loop back to normal when finish.

To trim and span by using hart: go to 1 device setup -> 2 diag/Service -> 3 Calibration -> 3 Pres sensor trim -> 1 Pres Zero trip and follow instruction to perform zero sensor and go to 1 device setup -> 2 diag/Service -> 3 Calibration -> 1 Re-range -> 1 keypad input to perform span adjustment until the result in tolerance accepted.

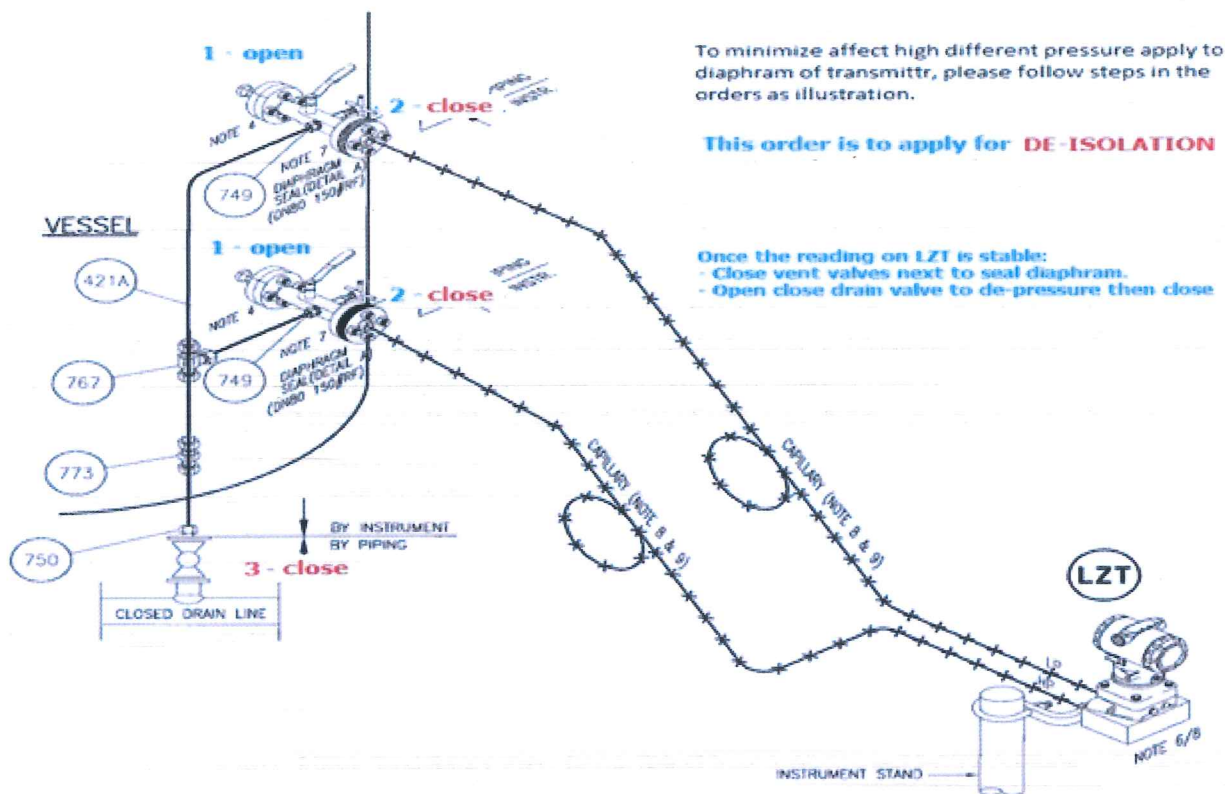
g. Use HART 475 go to 1 device setup -> 2 diag/Service -> 2 loop test -> 3 other to force the loop output to Alarm Setpoints and check Alarm on HMI.

h. Put the loop back to normal when finish. After finish disconnect HART communicator.

CALIBRATION SEQUENCES USE HART COMMUNICATOR OF LEVEL TRANSMITTER



- i. De-isolation as instruction figure below, put the transmitter back to normal, call CRE to remove-MOS, clear workplace.



6.2.3 Post Work Instruction

- Ensure equipment is left in a safe condition and the work area left clean and tidy.
- Double check and confirm MOS/INHIBIT is removed.
- On completion, close the Permit to Work and Work Order in Maximo with detail information.
- Finish job, make final report and record any defect to PM record form BD-OPS-MNR-I-3304/ F1

7.0 RECORDS

No	Name of record	Code of Form	Responsibility of keeping	Time of keeping
1	Safety Level Transmitter (DP) with Diaphragm Seal	BD-OPS-MNR-I-3304/F1	E&I Dept.	2 years

8.0 ATTACHMENTS

	PETROVIETNAM BRANCH BIEN DONG PETROLEUM OPERATING COMPANY MAINTENANCE RECORD	BDOPSMNRP3304/F1			
System name: _____ System: _____ Location: _____ Maintenance Type: <u>6M PM</u> Follow Instruction: <u>BD-OPS-MNR-I-3110</u> Safety Level Transmitter with Diaphragm Seal Maintenance Instruction.					
1. Prepare before maintenance					
☐ Prepare tools, gas tester and safety equipment ☐ Hazards identified and Precautions to be taken ☐ Hot work permit/ Cold work permit/WO No: _____ Others preparations: _____ ☐ Isolate, put devices in the MOS mode ☐ Double check and confirm with CREs about MOS/INHIBIT mode are applied before carry out every testing					
2. Test Equipment used for maintenance					
Testing Equipment	E&I Tool No	Model	S/n:	Manufacture	Calibrate Due to
3. Maintenance					
☐ Visually inspect the components for any signs of damage, corrosion ☐ Clean control panel, Marshalling cabinet ☐ Check tag name ☐ Put grease on all bolts, earth connector and cover nipple of devices ☐ Ex rating inspection Other jobs: _____ ☐ Check Earthing connection status 0.1 - 1ohm ☐ Check junction box and terminal ☐ Check instrument air supply, air regulator for SDVs, BDVs and control valves ☐ Check leakage air on SDVs, BDVs, and control valves					

4. Results										
No	Tag Name	Model	S/N	Description	Function	As found	As left	DCS/HMI	Result	Remark
1										
2										
3										
5. Check the system after maintenance										
☐ Check the completeness of the system ☐ De-isolation, cancel MOS of the equipment										
☐ Confirm with CRE F&G device back to normal after testing										
Other jobs: _____										
6. System's Status										
☐ Good, ready for operation ☐ Can temporary operate ☐ Need to fix immediately, not allow to operate										
Total execution time maintenance: _____ Hours Manpower: _____ Persons										
7. Consumable Materials										
9. Notes, suggestions, recommendations:										
		Position		Full Name		Signature		Date		
Completed By										
Verified By										
Approved By										